Indonesian Food Image Classification

A Deep Learning System for Recognizing Traditional Indonesian Dishes from Images

Project Overview

This project is an image classification system for traditional Indonesian food, built on a Convolutional Neural Network (CNN) using a transfer learning approach. The model was trained to recognize 20 categories of Indonesian dishes, ranging from staple meals such as Nasi Goreng, Rendang, and Soto, to street-food snacks such as Cilok, Batagor, Karedok, Ketoprak, and Siomay. The trained model was then deployed as an interactive web application that lets users upload a photo of food and instantly receive a prediction of what dish it is.

The main motivation behind this project was to build an image recognition system relevant to a local context (Indonesian cuisine), while practicing the full end-to-end machine learning workflow: dataset preparation, model training, performance evaluation, and finally deployment into a publicly accessible application.

■ Try the Live Demo

https://wirafadhil-klasifikasi-makanan.hf.space/

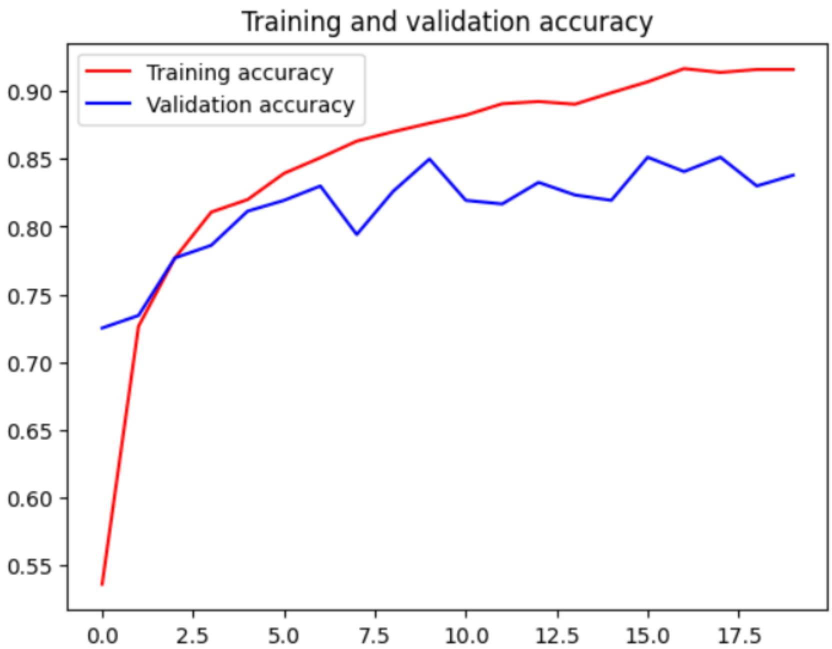
Click the link above to open the app directly (hosted on Hugging Face Spaces).

Technologies Used

- **Model Architecture:** Convolutional Neural Network (CNN) with transfer learning, using a pretrained backbone (e.g. EfficientNetB0 / MobileNetV2) for feature extraction, then fine-tuned on the Indonesian food dataset.
- **Dataset:** A collection of images across 20 Indonesian food classes, processed through preprocessing steps (resizing, normalization, augmentation) before entering the training pipeline.
- **Training & Evaluation:** The training process was monitored using accuracy and loss curves (training vs. validation) to detect overfitting, followed by in-depth evaluation using a confusion matrix and per-class accuracy.
- **Deployment:** The model was integrated into a simple web application (upload an image → get an instant prediction) and hosted publicly on Hugging Face Spaces so it can be accessed by anyone without installation.

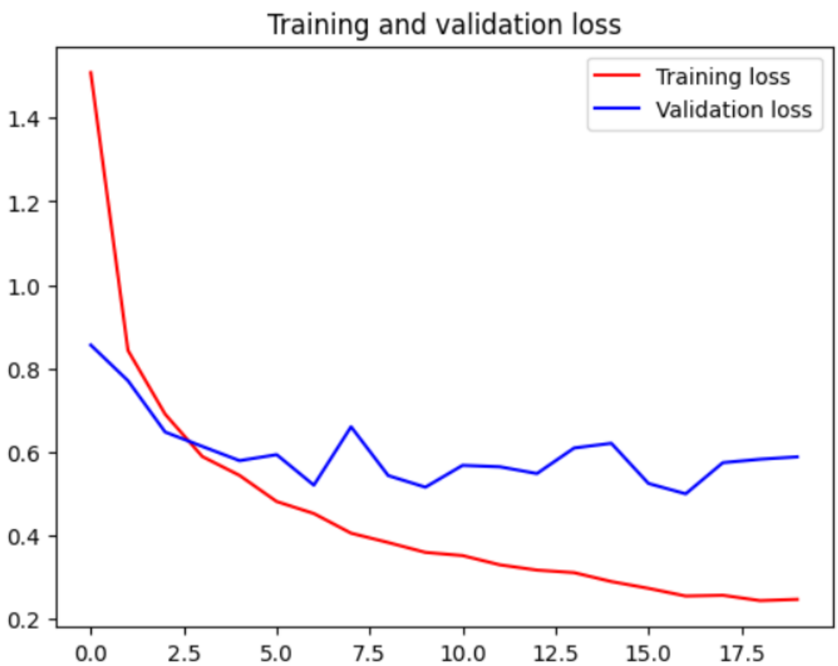
Model Training Process

The model was trained for 20 epochs. The accuracy curve shows a consistent improvement on both the training set (reaching around 91-92%) and the validation set (stabilizing around 82-85%), indicating that the model successfully learned the visual patterns of each food category without severe overfitting.



Training and validation accuracy over 20 training epochs.

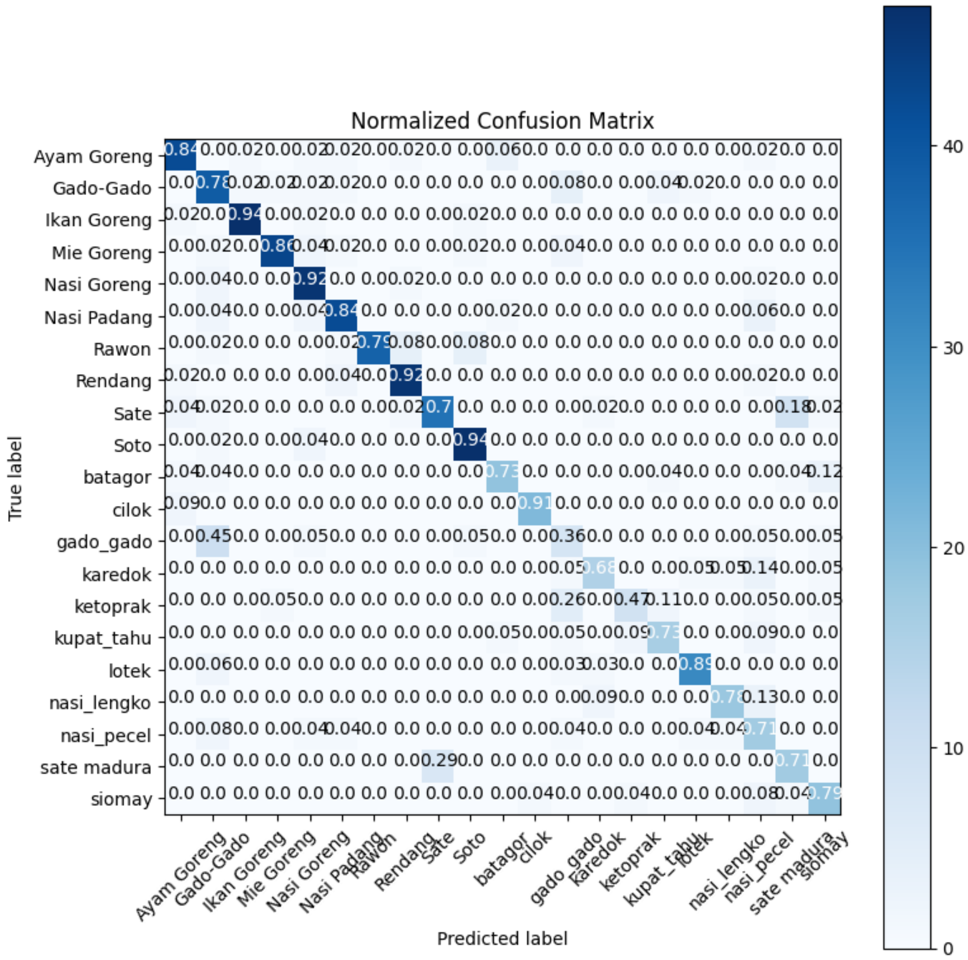
In line with this, the loss curve shows a sharp decline during the first few epochs before gradually leveling off. Validation loss fluctuates slightly but remains within a stable range, confirming that the model generalizes reasonably well to data it has not seen before.



Training and validation loss over the training process.

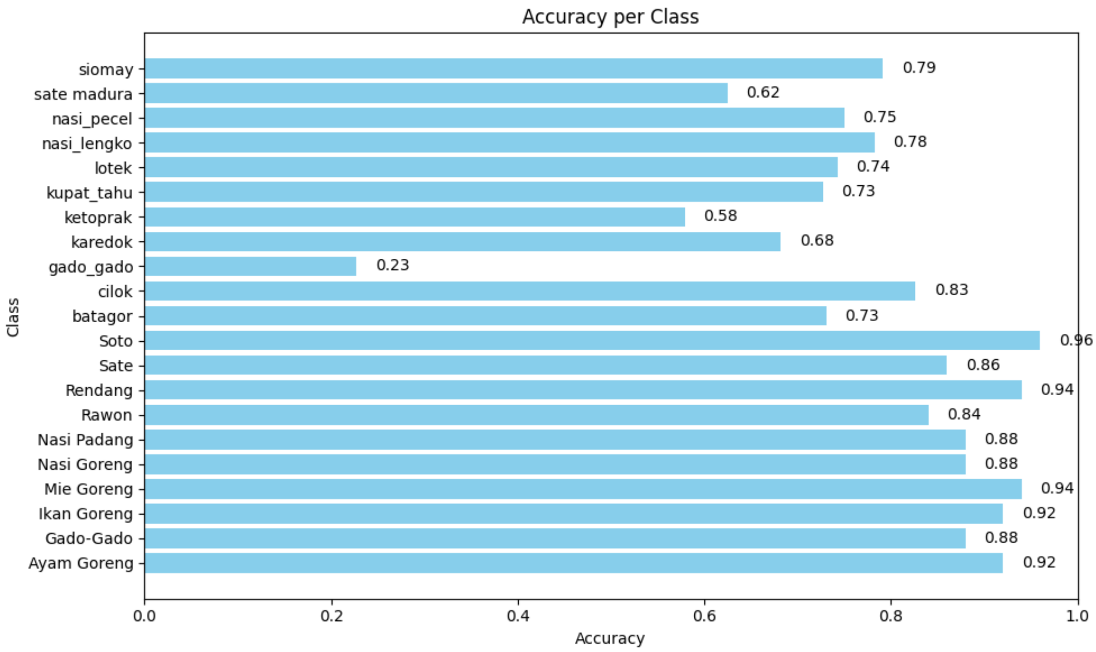
Model Performance Evaluation

To measure performance in more detail per category, a normalized confusion matrix was analyzed. Most staple-meal classes (such as Soto, Rendang, Ikan Goreng, and Nasi Goreng) were classified with high accuracy (above 90%). Meanwhile, several street-food classes that share strong visual similarity — for example Gado-Gado with Karedok, or Ketoprak with Kupat Tahu — show a higher rate of misclassification due to similar ingredients and sauce/broth appearance.



Normalized confusion matrix for the 20 food classes.

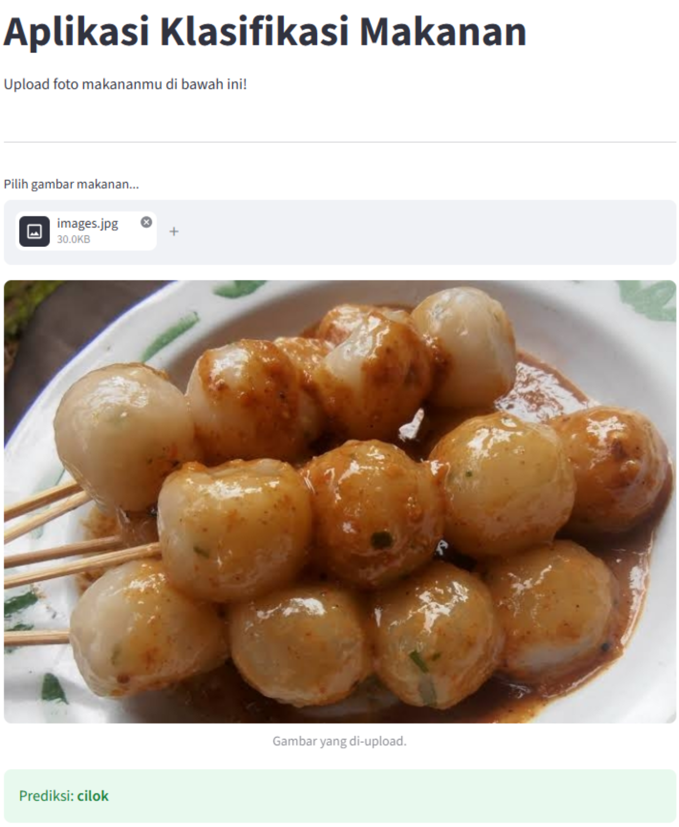
The per-class accuracy chart below makes this finding more quantitatively clear. Classes with distinctive and consistent visual appearance (Soto, Rendang, Sate, Mie Goreng) reach 94-96% accuracy, while classes with high presentation variation or strong inter-class similarity such as Gado-Gado (0.23), Ketoprak (0.58), and Karedok (0.68) stand out as the main challenge and a clear direction for future improvement — for instance by adding more training data or applying targeted augmentation for those specific classes.



Classification accuracy for each of the 20 food classes.

Implementation & App Demo

The trained model was then integrated into a web application called **"Food Classification App"**. Users simply upload a photo of food, and the system displays the image along with the predicted food label instantly. In the example below, an uploaded photo of skewered cilok was correctly predicted as **"cilok"**.



Application interface and an example prediction result.

The application is publicly deployed and can be accessed at: https://wirafadhil-klasifikasi-makanan.hf.space/

Key Achievements

- Built an image classification model for 20 Indonesian food classes, achieving an average validation accuracy above 80%.
- Conducted thorough model evaluation using a confusion matrix and per-class accuracy to understand the model's strengths and limitations.
- Successfully deployed the machine learning model as a functional, publicly accessible, real-time web application.
- Identified concrete challenges in classifying visually similar foods, providing a clear direction for further improvement.